

Figure Classification

COMPLETE LEARNING GUIDE

What You'll Learn

Figure classification (odd-figure-out) questions present a set of 4-5 figures where a clear majority share one specific, nameable property, and ask you to identify the single figure that breaks it. This sub-module goes deeper than the brief classification coverage in 'Pattern recognition' by working through the full range of properties non-verbal classification tests actually use — side count, symmetry, open vs. closed, rotation relationships within the set, shading/fill pattern, element count, and compound (two-property) rules — each with a fully worked set of figures and the exact reasoning that isolates the odd one.

1. Single-Property Classification: Side Count

The simplest classification rule groups figures purely by number of sides or number of line segments — scan all figures, tally each one's side count, and the figure with a side count that appears only once (while the others share a common count) is the odd one out.

- Five figures have side counts 5, 5, 6, 5, 5 (a pentagon, pentagon, hexagon, pentagon, pentagon). Four are pentagons (5 sides); the hexagon (6 sides) is the odd one out.

2. Symmetry-Based Classification

When all figures have the same side count, check the number of lines of symmetry each one has — figures can look structurally similar (same number of sides) yet differ in symmetry because their sides/angles aren't arranged evenly, so this property catches distractors that survive a pure side-count check.

- Four quadrilaterals all have 4 sides. Three are squares (4 lines of symmetry each); one is a generic (irregular) quadrilateral with no lines of symmetry at all despite also having 4 sides. The irregular quadrilateral is the odd one out, even though the side-count check alone wouldn't have caught it.

3. Open vs. Closed Figures

Some sets mix figures that form a fully closed boundary (every line connects back to form an enclosed region) with one figure that has a gap — a break in the outline where two ends don't meet — making it an open figure; this is checked by mentally tracing the full outline and confirming it returns to its starting point.

- Four star-like figures are shown. Three have their outlines fully closed (tracing the outline returns to the start). One has a small gap between two of its points where the lines don't quite meet, leaving it open. That open figure is the odd one out.

4. Shading and Fill Pattern

When shape and side count are identical across all figures, check the fill: is the figure fully shaded (solid black), unshaded (outline only), or partially shaded (e.g., only one section filled) — and if partial, whether the shaded portion is in a consistent relative position (like always the top half) across the majority.

- Four identical circles are divided into 4 equal quadrants. In three circles, the top-left quadrant is shaded. In the fourth, the top-right quadrant is shaded instead. All four have identical shape and an identical AMOUNT of shading (one quadrant), so only the POSITION of the shaded region reveals the odd one out.

5. Rotation Relationships Within the Set

In some sets, every figure is actually the SAME base figure rotated by different amounts — these aren't odd-shape questions but odd-rotation questions, where four figures are related to each other by clean rotation angles (like 0° , 90° , 180° , 270°) and the odd one is at an angle that breaks that clean set, or is a mirror image rather than a rotation.

- Four arrow-shaped figures point up, right, down, and left respectively (rotations of 0° , 90° , 180° , 270° of the same arrow). A fifth figure looks like it points left too, but on close inspection it's the mirror image of the 'left' arrow (its internal notch is on the wrong side) rather than a true 90° -step rotation. That mirrored figure is the odd one out.

6. Element Count Classification

Figures built from repeated small elements (dots, small circles, short lines within a larger boundary) are classified by counting those elements — the outer shape may be identical across all figures, with only the internal element count differing for the odd one.

- Four identical pentagons each contain small dots inside them: 3, 3, 3, and 4 dots respectively. All four pentagons look identical from the outside; only the dot count reveals that the 4-dot pentagon is the odd one out.

7. Compound (Two-Property) Rules

The hardest classification sets require TWO properties to hold simultaneously for a figure to belong to the majority group (e.g., 'has 4 sides AND is fully shaded') — check each property independently across all figures first, since a figure can fail on just one of the two properties and still be the unique odd one out.

- Four figures: three are shaded triangles, one is an UNSHADED triangle, and separately one of the shaded ones is actually a shaded quadrilateral. Testing 'triangle' alone: only the quadrilateral fails, since all others (including the unshaded one) are triangles — the unshaded triangle merely fails the separate 'shaded' property. Since three figures share BOTH properties (triangle AND shaded) and each of the other two fails a different single property, the figure failing the more fundamental shape property — the shaded quadrilateral — is the intended odd

one out.

8. Verifying the Rule Against the Full Majority

After spotting a candidate rule from 2-3 figures, always check it against every remaining figure in the set before finalizing an answer — a rule that happens to fit 3 out of 5 figures by coincidence is not reliable; a genuine classification rule should cleanly fit 4 out of 5 (or however many are in the set minus one).

- A candidate rule 'all figures have exactly one curved side' fits figures 1, 2, and 3. Checking figure 4 shows it also has one curved side (rule holds, 4/5 fit). Figure 5 has zero curved sides (all straight) — confirmed as the odd one out, since the rule now cleanly covers a full majority of 4, not just 3.

Common Pitfalls

- Stopping at the first shared property noticed (like side count) without checking whether it's the property that actually isolates a unique odd figure.
- Confusing 'same side count, different symmetry' figures as identical because they look superficially similar.
- Missing a small gap that makes a figure open rather than closed, especially in cluttered or star-shaped figures.
- In shading questions, comparing only the AMOUNT shaded and missing that the POSITION of the shaded region is the actual distinguishing property.
- Treating a mirror-image figure as a valid rotation of the base figure in rotation-relationship sets.
- In compound-rule sets, finalizing an answer after checking only one of the two required properties.
- Accepting a candidate rule that fits only a bare 3-out-of-5 figures instead of verifying it cleanly covers a full 4-out-of-5 majority.

Key Takeaways

- Check properties in a deliberate order — side count, then symmetry, then open/closed, then shading, then element count — rather than settling on the first one noticed.
- A rule is only valid once verified against every figure in the set, not just the first two or three.
- Watch for figures that are mirror images rather than true rotations when a set is built from one rotated base figure.
- In shading-based sets, compare both the AMOUNT and the POSITION of shaded regions.
- Compound rules require checking two properties independently — a figure can be the odd one out by failing just one of them.
- A genuine classification rule should cleanly fit a full majority (all but one figure), not just a coincidental few.

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IndiaBix — Non-Verbal Reasoning: Classification

<https://www.indiabix.com/non-verbal-reasoning/classification/>

GeeksforGeeks — Non-Verbal Reasoning Questions and Answers

<https://www.geeksforgeeks.org/non-verbal-reasoning-questions-and-answers/>